

Lab 4 – Vibe Coding

GitHub Copilot Chat History — Defender Repair Lab

Source: VS Code GitHub Copilot Chat export (chat.json)

This PDF presents the user prompts and the substantive Copilot responses from the exported chat session. Internal tool/telemetry records are omitted for readability.

Task 1 — Fix draw_radar()

User prompt

I am working on a Python/Pygame Defender game.

Task 1: Fix the broken draw_radar() function.

The radar should display the positions of all relevant game-world objects correctly across the radar, regardless of the player's current position.

Currently, the radar blips incorrectly cluster around the player's position because the radar is using player-relative coordinates incorrectly.

Please:

1. Inspect the existing game.py code and understand how world coordinates, player position, radar position, and radar dimensions are represented.
2. Identify the exact cause of the radar bug.
3. Modify ONLY the code necessary to fix draw_radar().
4. Preserve the existing game behavior, controls, graphics, and other functions.
5. Do not implement any of the other README feature tasks yet.
6. Explain what you changed and why.
7. Provide the exact code changes needed.

Copilot response

, which centers objects around the player for the main viewport. That made radar positions shift with the camera instead of showing each object's absolute position in the wrapping world.

Replace this line:

```
```python
rx = self.screen_x(x) % VIEW_W
```
```

with:

```
```python
rx = (x % WORLD_W) / WORLD_W * VIEW_W
```
```

This maps each wrapped world x-coordinate onto the full radar width, independently of the player's position. The y mapping and all other game behavior are unchanged. I didn't run the validation command, as requested.

Code shown in the chat

```
rx = self.screen_x(x) % VIEW_W

rx = (x % WORLD_W) / WORLD_W * VIEW_W
```

Task 2 — Implement sky_color()

User prompt

Implement Task 2 only: sky_color(wave).

Inspect the existing game.py and README requirements before making the change.

Requirements:

- Implement the existing sky_color(wave) function.
- It must return an (r, g, b) RGB tuple for the current wave.
- The sky color should change with the wave number so that different waves have visibly different sky colors.
- Keep the returned RGB values valid for Pygame (0-255).
- Preserve the existing game structure.
- Do not modify draw(), Game, draw_radar(), or any other functions.
- Do not implement on_humanoid_rescued() or bonus_life_threshold() yet.
- Make only the minimal change required for Task 2.
- Explain the implementation after making the change.

Copilot response

using a six-color palette that cycles every six waves. Each color is a valid RGB tuple, and the rest of the game code is unchanged.

Validation passed: six consecutive waves produce distinct colors, all channels are within `0–255`, and

Task 3 — Implement on_humanoid_rescued()

User prompt

implement the chnage mentioned above inn game.py

Copilot response

renders a rising `+500` above the rescued humanoid for 900 ms, then stops showing it. I also corrected the humanoid draw loop's indentation so the rendering code runs. Tasks 1, 2, and 4 were left unchanged.

A runtime check confirmed the popup appears and expires, and

Task 4 — Implement bonus_life_threshold()

User prompt

Implement Task 4 only: bonus_life_threshold().

Inspect the existing game.py and README requirements carefully.

Requirements:

- Implement the existing bonus_life_threshold() function according to the README.
- Return the required score threshold for earning bonus lives.
- The existing Game.update() logic already handles awarding the extra life, so do not duplicate that logic.
- Do not modify Game.update().
- Do not modify draw_radar(), sky_color(), or on_humanoid_rescued().
- Make only the minimal change required for Task 4.
- Explain what value you selected and why it matches the README requirement.

Copilot response

already handles awarding a life at each crossed threshold, so its logic is unchanged.

Verified the function returns `10000` and

Chat Export

User prompt

Chat: Export Chat

Copilot response

I can't run **Chat: Export Chat** from this session. In VS Code, open the Command Palette with `Ctrl+Shift+P`, run **Chat: Export Chat**, then choose the export location and format.